Synthesis and Characterization of CdSe/ZnS Quantum Dots and Silver Nanoparticles

1. Introduction

Förster resonance energy transfer (FRET) is a phenomenon involving the non-radiative transfer of energy between a fluorophore and quencher due to dipole-dipole coupling. Notably, this effect scales to R^-6, making this an excellent way of detecting molecular interactions (ie, enzymatic cleavage, protein-protein interaction, etc). However, low quantum yield/efficiency, photobleaching, and low spectral tunability mean that small molecule fluorophores/quenchers have certain disadvantages with respect to FRET.

To address these issues, this project aimed to synthesize CdSe/ZnS core-shell quantum dots (CdSe/ZnS QDs) as high-efficiency fluorophores and silver nanoparticles (AgNPs) as the quencher. CdSe/ZnS QDs, nanocrystals of cadmium selenide with a zinc sulfide shell, were selected due to the high quantum yield (due to ZnS shell passivation), high photostability, and highly tunable fluorescence inherent to quantum dots of this type, while strong size-dependent optical absorption from AgNPs led us to hypothesize that AgNPs could quench CdSe/ZnS fluorescence by Förster resonance energy transfer.

Our project aimed to test the hypothesis that CdSe/ZnS QDs would exhibit strong fluorescence and λmax in the 600-630 nm range, and AgNPs would exhibit strong quenching due to proximity-based energy transfer with QDs. To do so, we synthesized both nanomaterials and characterized them with TEM and UV/VIS spectrometry. Due to time constraints and solubility challenges with our existing quantum dots, quenching experiments were conducted with carboxyl-functionalized ITK 605 CdSe/ZnS quantum dots from Invitrogen as fluorophore.

2. Methods

CdSe Synthesis: We synthesized the CdSe/ZnS QDs by a modified hot-injection protocol (Source 1). The CdSe core was, in brief, synthesized by hot injection of 2 ml trioctylphosphine (TOP) selenide into a 270 C solution of cadmium stearate (0.2 mmol CdO dissolved in 0.8 mmol stearic acid at 220 C) and 4.9 g octadecylamine (ODA) dissolved in 8 ml octadecene (ODE). The CdSe nanocrystals were allowed to nucleate for 5 minutes at 250 C, after which the CdSe QDs were then extracted via hexanes, and purified via ethanol/chloroform-based procedure.

Precursor synthesis: Zn precursor (0.1M) was synthesized by dissolving 2 mmol ZnO in 16 mmol oleic acid and 15 ml ODE, while S-ODE (0.1M) was synthesized by dissolving sulfur in ODE. Calculated amounts were injected for SILAR.

ZnS coating via SILAR: ZnS coating was achieved by selective ionic layer absorption and reaction (SILAR). For SILAR, 7.7*10^-5 mmol of CdSe QDs, dissolved in hexane(2.5 ml), ODA(0.8 grams) and ODE (4 ml) were heated to 140 C to remove

hexanes. After injection of TOP (0.73 ml), the quantum dots were heated at 200 C for 30 minutes, followed by the injection of 0.61 ml of Zn precursor. The mixture was then maintained at 200 C for 30 minutes, after which S-ODE was injected, and the ZnS layer was allowed to grow for 60 minutes at 220 C, followed by cooling to 140 C.

For all consecutive layers, the reaction mixture was heated to 180 C, TOP (0.73 ml) was injected, and a calculated amount of Zn and S-ODE were injected, at a 10 minute interval, and then cooled to 140 C. This cycle can be repeated as many times as needed to get the corresponding number of layers (2 in our case).

AgNP synthesis: AgNPs were synthesized via the following procedure(Source 2). A solution of 10 ml AgNO3 (1.0 mM) was added dropwise into 30 ml of ice-cold NaBH4 solution (2.0 mM) under vigorous stirring. The resulting nanoparticles were then stabilized by adding 1 ml 0.3% PVP (MW = 40,000), which replaced the borohydride ions. The solution was centrifuged at 21000 rcf for 3.5 hours, and the pellet (containing AgNPs) was resuspended in water.

Quenching experiment: Quenching was achieved with AgNPs and water-solubilized CdSe/ZnS nanoparticles (Qdot™ 605 ITK™, Invitrogen/Thermo-Fisher), used due to time constraints and solubility issues. A QD stock solution was made by mixing 1 mL PBS (pH = 7) and 1 µL QDs, then mixing 600 µL of that solution with 600 µL of PBS (pH = 7). 100 µl of QDs, plus varying amounts of as-synthesized AgNPs diluted to 50 percent of original concentration with PBS (pH = 7). Fluorescence was measured by kinetic spectrophotometry (Spectramax I3X, excitation at 350 nm and 300 nm, fluorescence at 605 nm measured) to determine the quenching effect from AgNPs.

Characterization: For all three particle types, TEM was used to measure particle size and morphology. UV/VIS spectroscopy (Spectramax I3X) was used to analyze optical properties. For fluorescence measurements, samples were excited at 400 nm. A Spectramax I3X was also used for quenching analysis.

3. Results/Data



Top 3, left to right: CdSe/ZnS and CdSe fluorescence combined, CdSe/ZnS and CdSe absorbance combined, AgNO absorbance. Fluorescence was not measured for AgNPs.

Bottom 3, left to right. Photographs and TEM images of samples, quenching when excited at 300 nm, and when excited at 350 nm. Left to right for far-left: CdSe, CdSe/ZnS, and AgNP. Top is TEM images, at 20 nm, 50 nm, and 100 nm respectively.

4. Discussion

TEM and UV/VIS spectral data confirmed the successful synthesis of AgNPs and CdSe/ZnS quantum dots. The CdSe cores were found to have a size of ~5.41 nm and λmax = 600 nm, while the CdSe/ZnS core-shell QDs were found to have a size of ~6.87 nm with λmax redshifted to ~625 nm, consistent with successful shell formation. The AgNPs were found to be around 10.6 nm and absorb at 400 nm.

Due to time constraints, we were unable to get the CdSe/ZnS QDs water solubilized or measure quantum yield. However, we were able to confirm that our silver nanoparticles quenched water-solubilized ITK 605 CdSe/ZnS QDs with λmax = 605 nm. The results supported our hypothesis that a, CdSe/ZnS QDs are useful visible-range fluorophores, and b, AgNPs can serve as quenchers for quantum dot fluorescence.

Our future research directions include solubilizing our CdSe/ZnS QDs in water, making accurate quantum yield measurements, testing quenching with our own quantum dots, more accurate FRET measurements, and bioconjugating AgNPs and CdSe/ZnS quantum dots.